

A Mini-Project 1 Report on

IoT-Based Non-Destructive Invasive Species Suppression System

By

Kopunamoni Pranitha(24251A05E6)

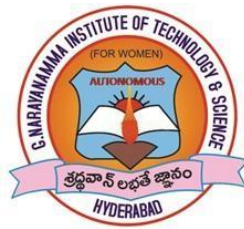
Maudshetty Sneha(24251A05F1)

Palaparthi Vaishnavi(24251A05F6)

under the guidance of

Dr. Jayashree S.Patil

Associate Professor



Department of Computer Science & Engineering
G. Narayanamma Institute of Technology & Science
AUTONOMOUS (For Women)
Shaikpet, Hyderabad 500 104.

Affiliated to
Jawaharlal Nehru Technological University Hyderabad
Hyderabad – 500 085
June, 2026

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Submitted to the Department of Computer Science & Engineering, GNITS in the partial fulfillment of the academic requirement for the award of B.Tech (CSE) under JNTUH

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G. Narayanamma Institute of Technology & Science

AUTONOMOUS (For Women)

Shaikpet, Hyderabad – 500 104.

Department of Computer Science & Engineering



Certificate

This is to certify that the Mini-Project 1 report on “**IoT-Based Non-Destructive Invasive Species Suppression System**” is a bonafide work carried out by **Kopunamoni Pranitha(24251A05E6)**, **Maudshetty Sneha(24251A05F1)**, **Palaparthi Vaishnavi(24251A05F6)** in the partial fulfillment for the award of B.Tech degree in Computer Science & Engineering, G. Narayanamma Institute of Technology & Science, Shaikpet, Hyderabad, affiliated to Jawaharlal Nehru Technological University, Hyderabad under our guidance and supervision.

The results embodied in the project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Internal Guide
Dr. Jayashree S.Patil
Associate Professor
Department of CSE

Head of the Department
Dr. A.Sharada
Professor & Head of
Department of CSE

G. Narayanamma Institute of Technology & Science

AUTONOMOUS (For Women)

Shaikpet, Hyderabad – 500 104.

Department of Computer Science & Engineering

IoT enabled CoE



Certificate

This is to certify that Kopunamoni Pranitha(24251A05E6), Maudshetty Sneha(24251A05F1), Palaparthi Vaishnavi(24251A05F6) of II B.Tech, have successfully completed project work in “IoT enabled CoE”,CSE. The project titled IoT-Based Non-Destructive Invasive Species Suppression System that is being submitted in partial fulfillment for the award of B.Tech, Computer Science and Engineering , G.Narayanamma Institute of Technology & Science affiliated to JNTUH, Hyderabad is a record of bonafide work carried out by them in our guidance and supervision.

**Internal Guide
Dr.Jayashree S.Patil
Associate Professor
Department of CSE**

**CoE Incharge
Dr. T. Rajesh,
Assistant Professor,
Department of CSE.**

**Head of the Department
Dr. A. Sharada, Professor
and Head of Department of CSE**

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Kopunamoni Pranitha-24251A05E6
Maudshetty Sneha-24251A05F1
Palaparthi Vaishnavi-24251A05F6

ABSTRACT

Excessive algae growth and deteriorating water quality pose significant threats to aquatic ecosystems by reducing dissolved oxygen levels, increasing turbidity, and disrupting ecological balance. Conventional monitoring and control methods, such as manual inspection and laboratory analysis, are often costly, time-consuming, and environmentally intrusive. To address these challenges, this project proposes an IoT-based water quality and algae bloom monitoring system using an ESP32 microcontroller and multiple sensors, including turbidity, temperature, light intensity, gas, and water level sensors.

Sensor data are continuously collected and compared with predefined threshold values to assess algae bloom risk. Based on the detected conditions, the system automatically activates appropriate actuators to control environmental factors and reduce bloom formation. To improve energy efficiency, sensors operate at scheduled intervals. An OLED display provides real-time status updates, while a buzzer and WhatsApp notifications alert users when intervention is required. The system also recommends preventive measures such as shade-net deployment to regulate light exposure. The proposed solution is simple, cost-effective, and easy to implement, offering real-time monitoring and automated response without relying on Artificial Intelligence (AI) or cloud-based processing. It supports sustainable water resource management and contributes to the protection of aquatic ecosystems.

Keywords: Internet of Things (IoT), Algae Bloom Monitoring, Water Quality Monitoring, ESP32, Aquatic Ecosystem, Environmental Monitoring, Automated Suppression System, Sustainable Technology.

Sustainable Development Goals (SDGs): SDG 14 – Life Below Water, SDG 15 – Life on Land.

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1. INTRODUCTION

Freshwater habitats (lakes, ponds, creeks, wetlands and reservoirs) are some of our most important sources of land-based life, as well as aid in preventing pathogens from entering freshwater (through beneficial microorganisms working together to provide a balanced habitat). These habitats provide water for agricultural, industrial and domestic use. They contribute to wetland plant species; Therefore, they help produce and maintain biodiversity and regulate climate. Freshwater ecological systems contain fish, plants, microorganisms and other species that help keep them stable [23], [25].

However, due to the degradation or destruction of our freshwater habitats, what is left in our freshwater habitats can have a detrimental effect on both ecosystems and human populations. Until now, we have not encountered any major challenges in relation to freshwater resources because of rapid growth of urban cities, increased industrial production, the increasing population, all made worse by poor waste disposal [23], [22].

Untreated sewage, chemicals from farms, and the accumulated garbage create pollution in water. Procedural processes also cause uncontrolled growing of foreign or non-native species known as invasive species. Invasive species invade new places and usually proliferate rapidly and therefore to dominate an area by outcompeting native plants for nutrients (soil), sunlight, oxygen, carbon dioxide, water, and space. Invasive species that are water-dependent can also block light, decrease the oxygen levels, alter the flow of water, and disrupt the local food web. The introduction of non-native organisms can often lead to degraded water quality, increased fish mortality, reduced biodiversity, and higher management costs [22], [5].

It is generally accepted that traditional methods for controlling invasive species involve physical removal, mechanical cleaning, and/or chemical application. While the methods may work quickly, they also tend to be expensive, labour-intensive, and may result in inadvertent damage to native organisms or their habitats [5].

Technology of today provides smarter and sustainable methods to manage our ecosystems and the Internet of Things (IoT) has proven to be an effective method for doing so. The Internet of Things (IoT) can be defined as a collection of sensors and controllers connected through several means of communication capable of continuously collecting and sharing

data [2], [12].

An IoT-Based Non-Destructive Invasive Species Suppression System was designed to assist with early detection and environmentally friendly prevention of invasive species through real-time monitoring of habitat characteristics using an ESP32 microcontroller and multiple types of sensors including turbidity, temperature, light intensity, gas and water level. Data from these sensors provides insight into ecological imbalances and/or invasive growth allowing for preventive measures (e.g., controlled aeration or circulation of water, adjusting light levels) instead of destructive actions (e.g., poison or other types of toxic substances). Additionally, alerts can be generated if part of the monitored habitat reaches thresholds defined within the system, creating an environment conducive to preventing the establishment of invasive species and promoting the establishment of native species [2], [12], [1].

Overall, this system employs technology responsibly and enhances protection for aquatic ecosystems by supporting sustainable development. Thus, continuous monitoring, timely decision-making, and non-destructive interventions may greatly increase the protection of aquatic ecosystems and provide support to achieve sustainable development goals through environmentally responsible technology [12], [13].

1.1 Background of the Study

Freshwater ecosystems like ponds, rivers, and lakes are immensely critical to life on Earth since they serve as important sources of drinking water for humans, support agricultural production, and are home to many different types of species [23], [25]. They have also gravely degraded due to anthropogenic pressures such as urbanization, industrial expansion, and population growth [23], [22]. Anticipated increases in the number of people inhabiting peri-urban areas exert considerable pressure on freshwater ecosystems; this pressure has multiplied over time through increased agricultural runoff, wastewater discharge, and pollution [22]. Furthermore, there is the challenge of invasive species, which often unpredictably and rapidly populate native species habitats [5]. Invasive species can ultimately have detrimental effects on native plants and animal populations, affect habitat and food web dynamics, and lessen overall ecosystem productivity [5]. Invasive freshwater plant species specifically shade out native

vegetation by blocking sunlight; they also outcompete other vegetation for nutrients due to their ability to assimilate nutrients quickly, which reduces the chances of native vegetation surviving [5]. Unfortunately, in order to monitor changes in freshwater ecosystems, managers often rely on manual techniques, which require significant amounts of time, capital, and resources to implement [2].

Unfortunately, many times an ecosystem goes wrong before it is possible to identify and address the underlying causes. Often, by the time a problem becomes evident, remediation becomes more costly and complex. Therefore, there is a growing need to utilize technologies such as the Internet of Things (IoT) to make environmental management more responsive [12]. In this study, a framework is proposed to monitor ecological indicators and identify potential environmental threats before they escalate. The proposed approach combines scientific knowledge and engineering principles to support aquatic ecosystem protection [12], [1]. When a body of water loses its transparency due to increased sedimentation, certain aquatic plants may grow under altered lighting conditions [13]. The reduction of natural shading can disrupt ecological balance and significantly reduce dissolved oxygen levels due to limited natural circulation [13]. Prolonged exposure to such conditions may result in stress or mortality of aquatic organisms.

As sediment concentrations remain high for extended periods, ecosystem services may become degraded or lost entirely [13]. Traditional monitoring methods, such as manual sampling, are often unable to provide the real-time information required for prompt intervention, potentially leading to irreversible environmental damage before corrective measures can be implemented [2].

By utilizing IoT technology, which provides extensive connectivity through real-time and internet-connected sensors, continuous environmental monitoring can be achieved [12], [1]. Such a framework enables the early detection of conditions that may promote invasive species establishment and allows preventive actions such as controlled aeration and environmental regulation to be implemented at an early stage [12], [1]. Ultimately, this approach can improve ecosystem health, enhance water quality, and support sustainable habitat conditions for aquatic organisms rather than merely reacting after environmental degradation has occurred [12], [13].

1.2 Problem Statement

Freshwater ecosystems tend to have a low depth-to-surface area ratio (generally deeper than wide), which means they are sensitive to changes in their surrounding environment [23], [25]. Small changes to parameters, such as increased turbidity, elevated water temperature, and increased nutrient levels, can have a dramatic effect on ecological processes [23], [2]. For example, increases in algal growth, reductions in dissolved oxygen levels, and declines in native fish populations can result from changes in these parameters [23], [2]. A leading cause of continued ecological degradation is the limited capability of conventional monitoring systems, which often provide only post-event analysis of major environmental changes after the event has already occurred [2], [12]. By that time, environmental damage may be extensive, making recovery costly and difficult. Furthermore, conventional monitoring systems rely heavily on manual sampling and laboratory analysis to quantify environmental conditions. Although such methods can provide accurate measurements, they are often time-consuming and delay the detection of environmental changes [2].

Algal blooms are particularly evident in large Indian urban centres such as Hussain Sagar Lake in Hyderabad and Ulsoor Lake in Bengaluru [22]. Algal blooms can often be identified by their greenish appearance and unpleasant odour; however, visible signs generally appear only after significant ecological degradation has already occurred. The common response involves chemical treatment and mechanical removal, both of which may adversely affect native aquatic organisms, including fish, aquatic plants, and microorganisms that play essential roles in maintaining ecosystem stability [23], [5]. Chemical treatment can also alter the natural chemistry of the water body, potentially reducing biodiversity and increasing ecosystem vulnerability to future disturbances [5].

Cost is another factor limiting the widespread adoption of advanced monitoring methods. Laboratory-based analyses require trained personnel, specialized instruments, and sustained financial support [2]. Consequently, such approaches are difficult to implement on a large scale, particularly in rural regions and small community water bodies. Nevertheless, these smaller water bodies remain important resources for agriculture and wildlife conservation [23], [25]. In addition, continuous monitoring of small aquatic ecosystems remains largely unavailable [2], [12].

All three indicators—water turbidity, dissolved oxygen levels, and gas concentrations—can change rapidly [2], [13]. If monitoring is conducted infrequently, such as weekly or

monthly, environmental degradation may occur before corrective actions can be implemented [2], [12].

Therefore, the reasons for the identified challenge are as follows:

1. Continuous monitoring is not currently available. Existing monitoring approaches generally capture environmental changes only at periodic intervals [2], [12].
2. There is a significant delay between field sample collection and laboratory analysis, allowing ecological degradation to continue before results become available [2].
3. Existing intervention methods can negatively affect both invasive and native species [5].
4. The high cost of conventional water-quality monitoring technologies limits their adoption in rural areas and small community water bodies [2], [1].

1.3 Existing Systems

Traditionally, water quality monitoring has relied primarily on physical samples collected from a specific location at a particular point in time, enabling periodic assessment of water quality [2]. Although physical sampling provides information regarding pollution levels, it does not adequately capture the dynamic nature of aquatic environments or account for the speed and extent of pollutant dispersion after release [2], [13]. Furthermore, increasing urbanization and environmental variability necessitate continuous and near real-time monitoring approaches [12], [13]. Consequently, several limitations of existing monitoring methods have been identified.

1. Manual Sampling:

Manual sampling measures environmental parameters only at predetermined intervals and may therefore fail to detect sudden changes in water quality [2]. In contrast, the proposed framework supports continuous monitoring, enabling rapid detection of environmental variations [12], [1]. Traditional monitoring approaches depend heavily on human involvement, which may introduce inconsistencies and operational inefficiencies. The proposed framework automates both data acquisition and processing, thereby improving reliability and consistency [12], [1]. Additionally, manually collected data often require further processing before analysis and dissemination. The proposed

framework enables real-time digital data acquisition and monitoring [12], [1]. Conventional monitoring is frequently limited to specific sampling locations and may not adequately represent larger aquatic environments. Sensor-based monitoring improves spatial coverage and environmental observation capabilities [12]. Moreover, frequent field inspections increase labour costs and reduce scalability, whereas automated monitoring reduces the need for routine site visits [12], [1].

2. Laboratory Analysis:

Laboratory-based approaches and chemical treatment strategies generally require the presence of observable contamination before intervention can occur. Such methods may adversely affect aquatic organisms and disrupt ecological balance [5]. Excessive dependence on chemical treatment may contribute to long-term deterioration of water quality and often provides limited information regarding the origin of pollution sources [5]. In contrast, the proposed framework provides continuous environmental monitoring and trend analysis, enabling the identification of potential environmental changes at an earlier stage and supporting more effective intervention strategies [12], [1].

3. Chemical Treatment:

Chemical agents such as algaecides and chlorine are commonly used to suppress algal blooms [5]. Although these treatments may provide temporary improvements in water appearance, they can alter the natural chemical composition of water and negatively affect surrounding aquatic organisms [5]. Repeated chemical intervention may reduce biodiversity and increase ecosystem vulnerability to future environmental disturbances [23], [5].

4. Mechanical Treatment:

Mechanical removal and physical skimming are commonly used methods for removing algae from water bodies [5]. However, these approaches are resource-intensive, expensive, and often require repeated implementation [5], [2]. Furthermore, they address only the visible symptoms of algal blooms rather than the environmental conditions that contribute to their development, making future recurrence likely [5], [2].

As shown in Table 1.1, conventional algae monitoring and removal methods often suffer from limitations such as high operational costs, delayed response times, and temporary effectiveness. Existing approaches are generally labour-intensive and lack the capability for continuous real-time environmental monitoring[2],[12].

Table 1.1: Conventional Methods for Algae Monitoring and Removal in Water bodies along with their Limitations.

Method	Description	Limitations
Manual Sampling	Collecting water samples periodically	Time-consuming, delayed results
Laboratory Analysis	Testing turbidity, pH, oxygen levels	Expensive, not continuous
Chemical Treatment	Adding algaecides or chlorine	Harmful to native species
Mechanical Removal	Physical cleaning of algae	Labor-intensive, temporary.

The table presents the conventional methods used for algae monitoring and removal in water bodies, along with their major drawbacks. Manual sampling involves collecting water samples at regular intervals and analyzing them, but this process is time-consuming and does not provide immediate information about changing water conditions. Laboratory analysis is another common method where water parameters such as turbidity, pH, and oxygen levels are tested. Although it provides accurate results, it requires specialized equipment, skilled personnel, and is expensive, making continuous monitoring difficult.

For controlling algae growth, traditional methods such as chemical treatment and mechanical removal are widely used. Chemical treatments involve adding substances like algaecides or chlorine to eliminate algae, but these chemicals can also affect fish and other beneficial aquatic organisms. Mechanical removal physically cleans algae from the water, but it requires significant human effort and provides only temporary results. These limitations highlight the need for a smarter, continuous, and eco-friendly monitoring solution, such as an IoT-based aquatic monitoring system.

1.4 Drawbacks of Existing Systems

While many of the regular monitoring techniques used today have been utilized for the past couple of decades, they still present a myriad of serious issues that in turn have created barriers that make them less than perfect in today's ecosystem management today. It is not just technical issues; ecological, economic and societal issues also present substantially different types of challenges to achieving real and long lasting success through effective ecosystem management. When this perfect storm occurs, it creates obstacles that make it very difficult to obtain sustainable outcomes.

Lack of real-time monitoring

Most of the examining methods used today only use samples from the field taken periodically, and since those samples need to be tested in laboratories after the fact, typically, water quality conditions will be checked only once a week, perhaps once every two weeks or longer. This is in part because of the fact that the characteristics of aquatic ecosystems can vary constantly over short periods of time; continually monitoring an ecosystem can identify changes (like changes to turbidity, gas concentration or oxygen concentration) before those changes become irreversible. Conversely, if sampling takes place and is tested on a weekly/monthly basis, then an organization will likely not have time to act quickly, leading to a potentially negative results.

Slow Detection and response

The chain of events regarding lab results is very slow even though there really isn't a question of how trustworthy the lab results are. Laboratory samples must be obtained and taken to the appropriate location for analysis. Once they are analysed, the results won't be available for many hours or days afterwards. Therefore, by the time the proper authorities use the information from the laboratory to take the actions required to restore the balance ecosystems, it is possible that the entire ecosystem has already been altered. This can be further complicated in instances where there are algal blooms. In these instances, the algal blooms can quickly spread to many areas and, in many instances, they can smother all native species that are adjacent to them. Fish kills and odour issues associated with Bellandur Lake in Bengaluru have also been well documented and are due primarily to slow detection and slow response time. These responses have impacts both on biodiversity loss, as well as the well-being of humans.

Harmful interventions

Harmful methods are being used; chemicals (such as algaecides and chlorine) may seem to suppress an algal bloom in the short term -- however, they alter the water chemistry in a manner detrimental to native species; biodiversity decreases, decreases the resilience of the ecosystem, and eventually creates a system that becomes more susceptible to the next outbreak. Physical cleaning through manual removal methods have been used as well, but require substantial labor and provide only temporary reprieve. Furthermore, they do not address the major root cause of excessive nutrients and a lack of adequate monitoring will continue to create these repeated outbreaks.

Very high operating cost

Lab-based monitoring of the water environment relies on specialized equipment, trained personnel, and continual costs associated with the method as well as additional costs associated with treatment chemicals and mechanical removal of pollutant sources (e.g., tires, bottles). All of these factors make lab-based monitoring impractical for rural communities and smaller water bodies, both of which are critical for agricultural and biological diversity. Low-cost monitoring will allow for the identification of pollution in these ecosystems before large-scale damage occurs.

1.5 Proposed System

In summary, what has been created is an intelligent, IoT-based (internet of things) monitoring system -- designed to monitor the conditions of various water sources -- which uses existing methods of water quality monitoring as an information source. This system is considered to be proactive, as it continuously runs and uses fewer resources (i.e. chemical treatments) than traditional methods to process and share water-quality data (e.g. water-quality testing). This system uses multiple sensors connected to an ESP32 microcontroller that monitor multiple environmental variables (e.g. turbidity, temperature, light intensity, gas concentrations and water levels). The information that is collected by these various sensors is processed by the system in "real-time," and is rated for its risk of producing algae blooms using three different levels of risk (e.g. low, medium, high).

One of the benefits of this system is that it operates in a non-destructive manner. Instead of using chemical treatments, or mechanical approaches to remove algae directly, the system encourages corrective measures such as controlled aeration, shading with netting, and issuing environmental alerts. This allows for less harm to native species, and a

longer-lasting approach. When turbidity and gas levels exceed predefined thresholds, the system may recommend approximately 70% shade cover to limit light transmissions and control algal growth. At the same time, aerators can be turned on to increase dissolved oxygen levels, without directly impacting aquatic organisms. A second key benefit is the ability of this system to offer scalability and affordability. As it is based on the use of inexpensive sensors and microcontrollers, the overall costs associated with deploying the system are more easily achievable by rural communities and educational institutions. Additionally, multiple systems can be deployed simultaneously to multiple locations to facilitate continuous monitoring without incurring the costs of laboratory equipment or requiring highly skilled people.

In addition to sending out alerts in real-time through communication frameworks such as WhatsApp when a risk level is changing, the stakeholder will be alerted quickly and be able to plan for timely intervention before the situation gets negative. In terms of future enhancements to the proposed system, adding cloud integration, mobile application development, and possibly even solar-powered autonomous operation would make the system more adaptable and sustainable long term. The above information supports the concept of transitioning from reactive to proactive management of ecosystems through the use of IoT technologies combined with low-impact or environmentally-based interventions; thus, allowing for continual monitoring, early detection of problems, and sustainable, corrective measures to be taken. It remains cost-effective; it is a scalable solution; and it can be implemented in a variety of environments – including urban lakes and rural ponds.

1.6 Objectives

In freshwater ecosystem monitoring, the objectives are not just technical focuses, but also ecological and social. There are many people who are working towards promoting sustainable access and narrowing the gap between 'traditional' monitoring operations and 'new' IoT based monitoring operations; this process may be a bit uncoordinated initially. The main objectives of the project are:

- To continuously monitor key environmental parameters including turbidity, temperature, humidity, light intensity, gas concentration, and water level using integrated sensor modules.
- To develop a real-time risk assessment model that processes sensor data and

classifies ecosystem conditions into LOW, MEDIUM, and HIGH risk levels.

- To implement automated environmental control using relay-driven actuators such as water pumps, turbulence pumps, cooling fans, and buzzer alerts based on detected risk conditions.
- To provide real-time user notifications through a WhatsApp alert system for critical environmental conditions and early intervention.
- To develop a sustainable, non-chemical, and user-friendly solution for aquatic ecosystem protection through continuous monitoring, automated control, and early warning mechanisms.

The first objective is mostly about affordability. A lot of these setups depend on expensive laboratory instruments and trained personnel, so they are not really reachable for many rural communities. This project takes another path, it leans on low cost sensors plus microcontrollers, like ESP32, so ongoing monitoring becomes possible with a much smaller share of the budget. For example, a turbidity sensor can cost only a few hundred rupees, and it can deliver real time data that would otherwise need costly lab tests. Overall, the system remains scalable and budget friendly, and it can be placed across multiple water bodies with less trouble than before.

1.7 Methodology

This project wants you to know that the main purpose of its methodology is to demonstrate step-by-step how to design and build an IoT monitoring system, and then to put it into operation. This methodology is both systematic and flexible enough not to create an impression of randomness when working through the process workflow. The reason this is important is because every piece of data you gather from a sensor (such as an air quality sensor located outdoors, prior to being infected) can be gathered in a more ordered manner with a more cohesive flow of events from start to finish (e.g., prevention starting at the time you first became exposed through screening). The data gathered in this method will be considered more proactive because the entire set of events will be fast, predictable (i.e., it will be known/expected when to expect such events), and structured in a manner to allow them to be "knitted together" (not fragmented) than

traditional methods which tend to be reactive and disconnected.

The project follows something like this

Step 1: Sensor Data Collection

The project begins with sensors, which are the foundation of this project. The turbidity sensors check for the clarity of the water. Light sensors measure how much light is reaching the water surface, and temperature sensors continuously monitor the water temperature. Gas sensors monitor the presence of gas in the water, such as ammonia or carbon dioxide. Float sensors monitor how much water is present in that particular area. These sensors create an overall picture of the condition of the ecosystem; for example, if there is a sudden increase in turbidity and a steady increase in temperature, evidence may exist of a burgeoning algal bloom or the expansion of existing blooms.

Step 2: Signal Conditioning

The outputs of sensors are not always perfect, and there may be minor inaccuracies or noise amongst them. Therefore, making sure that the data is accurate and (more importantly) reliable to trust has to be accomplished. Complete signal conditioning consists of validating sensor functionality before removing erroneous readings and formatting signals to be recognizable by the microcontroller. If the signal conditioning is not included in this process, the resulting sensor data will provide incorrect risk labels. Signal conditioning and sensor data collection must be treated as a pair; otherwise, an entire Process of "collecting and conditioning of signal" would become unaligned.

Step 3: ESP32 Processing

The motherboard of the system is represented by the ESP32. The ESP will receive the values from the sensors, and determine if the values being given to the microcontroller appear to be acceptable. If there is a sudden rise in turbidity for example (a 600 NTU ++ value), and the gas values would appear to be different than normal as any other given time, then the ESP will signify to us with an indication such as "High Risk". The advantages of using the ESP32 as the brain of the system are that they are inexpensive, low-ish on power consumption, and can handle multiple sensor inputs at the same time - without getting out of control.

Step 4: Output Display

After processing, the ESP32 microcontroller transmits to the Arduino Serial Monitor

results. This lets users know in real-time what is happening with the device. Users can monitor the functionality of the sensors, and compare water quality trends. For example, if there is a turbidity measurement of 620 NTU, a measured temperature of approximately 29°C, and a designated Risk Level of Medium, we can interpret how the water behaves at that moment in time.

Step 5: Ecosystem Health Evaluation

The following process assesses the health of the ecosystem by creating a score using all of the measurements from the sensors. Each of the factors evaluated receives an individual score that is then summed to create a cumulative risk rating. The cumulative risk rating is then classified into three levels of risk: low risk, medium risk, and high risk. The use of scores allows the users to interpret multiple measurements at once, as well as to make a decision on an action without having to guess.

Step 6: Actuator Activation

Subsequent to identifying risk levels, we will engage supportive devices, including aerators, pumps and fans. The purpose of these devices is to help improve the condition of the system while being mindful of the environment. For example, if there is an extremely high risk level, we can use an aerator to increase the level of oxygen in the water, or we can use a shade net to reduce sunlight from reaching the system. These solutions are more beneficial than just using chemicals because they both enhance the health of living plants and animals as well as help prevent negative concerns from becoming the principal concern in the system.

Conclusion:

Fundamentally, the ecosystem management system will be proactive, reliable, and sustainable through the use of combined sensors, microcontrollers, monitoring tools and actuators. The combination of all four groups will create a workable way for land managers and landowners to improve the unmanaged conditions of their ecosystems. Each phase of the ecosystem management system will help identify early issues and make environmentally friendly choices. For this reason, the ecosystem management system is not only a concept but will also produce a significant, real-world result..

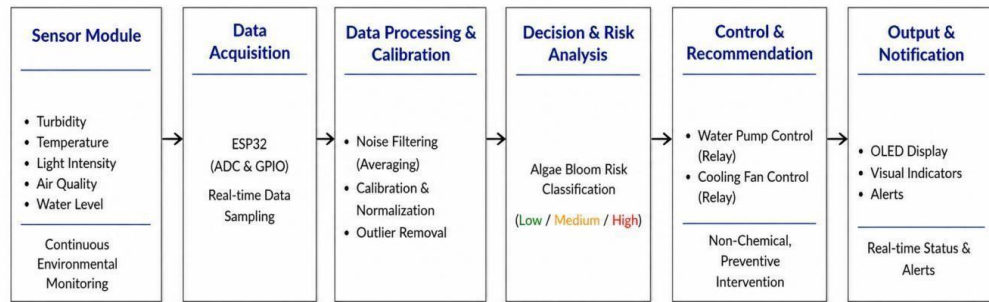


Figure 1.1: System Methodology

As shown in Figure 1.1, the proposed system methodology explains the complete workflow from environmental data sensing to real-time monitoring and automated control actions. From the figure, sensor data is processed and analyzed to classify algae bloom risk and provide suitable preventive recommendations and alerts.

2. LITERATURE SURVEY

Lokman et al. proposed an advanced Internet of Things (IoT) framework integrated with Artificial Intelligence (AI) to enable real-time monitoring of water quality in aquatic ecosystems. The framework incorporates multiple sensors to monitor important environmental indicators such as turbidity, pH, dissolved oxygen, and temperature, while Machine Learning (ML) algorithms are employed to detect anomalies and predict potential water quality contamination events. Sensor data are transmitted to cloud-based AI models that identify unusual environmental patterns and provide early warnings of possible pollution incidents [4], [20].

As a result, the monitoring process becomes not only more reliable but also more intelligent through the integration of AI and ML technologies. This enhances the system's ability to identify pollution trends and predict environmental impacts before significant damage occurs. The study demonstrated improved monitoring accuracy and decision-support capability compared to conventional monitoring approaches [4], [20].

Despite these advantages, the proposed framework has certain limitations. It requires substantial computational resources, extensive cloud-based processing, and a stable internet connection for effective operation. In addition, the implementation cost is relatively high because of the AI-intensive architecture. Consequently, such systems may not be suitable for rural or resource-constrained environments where internet connectivity and power availability are limited.

In contrast, the proposed work focuses on developing a cost-effective and energy-efficient IoT-based algae bloom risk monitoring framework using an ESP32 microcontroller. Instead of relying heavily on computationally intensive AI models, a threshold-based decision-making approach is employed to evaluate algae bloom risk. This enables the development of a low-cost and easy-to-deploy solution suitable for small-scale aquatic environments, particularly in rural areas where continuous cloud connectivity may not be available.

An IoT-based environmental monitoring framework was also developed by Wang and his colleagues using multiple interconnected sensors integrated with an ESP-based microcontroller. The framework utilized sensors to measure environmental parameters such

as temperature, turbidity, gas concentration, and light intensity. Their findings indicated that monitoring accuracy improved when data from multiple sensors were combined rather than relying on a single sensing parameter [18], [20], [7].

The results demonstrated that multi-sensor data fusion provides a more comprehensive representation of environmental conditions than individual sensor measurements. Furthermore, the sensors exhibited stable real-time performance with minimal maintenance requirements throughout the evaluation period [18], [20], [7].

However, there are certain limitations associated with the proposed framework. The experiments were conducted in relatively small and controlled environments, making it difficult to evaluate performance under large-scale aquatic ecosystem conditions or highly variable environmental scenarios. Furthermore, the framework primarily focused on monitoring and reporting environmental conditions without incorporating automated corrective actions.

The present study adopts the multi-sensor environmental monitoring concept introduced by Wang et al. by integrating turbidity, temperature, light intensity, gas, and water-level sensors into a unified monitoring platform. The proposed framework further extends this approach through algae bloom risk classification, automated relay-based actuator control, aeration mechanisms, and real-time WhatsApp alert notifications to support early intervention and prevention.

Xia and colleagues developed a machine learning-based methodology for predicting harmful algae outbreaks using environmental parameters such as turbidity, temperature, nutrient concentration, dissolved oxygen, and chlorophyll levels. Multiple machine learning algorithms were trained to identify patterns associated with algae bloom formation and forecast future environmental conditions. Their findings demonstrated that machine learning techniques could effectively capture complex relationships among environmental variables and provide predictive insights for proactive ecosystem management [10], [4].

Despite its effectiveness, the proposed model requires extensive historical environmental datasets for training and validation. Many rural and small-scale aquatic environments lack sufficient historical records, limiting the practical applicability of such approaches. Additionally, machine learning algorithms generally require greater computational

resources and energy consumption compared to simpler monitoring methods [10], [4].

The present work differs significantly from the approach proposed by Xia et al. Rather than relying on historical datasets and computationally intensive machine learning models, the proposed framework employs a real-time threshold-based risk scoring mechanism for algae bloom assessment. This approach reduces implementation cost, computational complexity, and energy consumption while enabling deployment in low-resource environments.

Rahman et al. developed a low-cost IoT-based water quality monitoring framework using an ESP32 microcontroller and sensors for measuring turbidity, temperature, and pH. Their objective was to demonstrate the feasibility of affordable real-time environmental monitoring for educational applications and small-scale field deployments. The prototype successfully monitored and transmitted water-quality data through wireless communication while maintaining a compact and cost-effective design [17], [18], [20].

However, the framework offered limited advanced functionality, such as intelligent risk assessment, automated intervention mechanisms, and remote notification services. As a result, its application remained largely experimental and focused primarily on observation rather than proactive ecosystem management. The present research extends the low-cost ESP32-based architecture proposed by Rahman et al. by incorporating additional functionalities, including multi-sensor integration, algae bloom risk classification, automated relay control, pump and aeration management, grouped sensing for power optimization, OLED-based real-time visualization, buzzer notifications, and WhatsApp-based remote alerts. Consequently, the proposed framework evolves beyond simple environmental monitoring and provides a more comprehensive solution for aquatic ecosystem management and algae bloom prevention.

From Table 2.1, it can be observed that existing aquatic ecosystem monitoring frameworks primarily rely on IoT, Artificial Intelligence, and Machine Learning technologies to improve monitoring accuracy and prediction capabilities. However, many of these approaches require high computational resources, cloud dependency, or extensive datasets. In contrast, the proposed project focuses on a simpler, cost-effective, and real-time monitoring framework based on threshold-driven decision logic, making it more suitable for practical deployment in resource-constrained environments.

Table 2.1: Summary of Existing systems on Continuous Aquatic Ecosystem Monitoring systems.

Author & Year	Method	Outcomes	Limitations	Relevance to Present Project
Lokman et al., 2025	AI-enabled IoT framework	Early contamination detection	Requires cloud & high computation	Present project avoids heavy computation
Wang et al., 2025	Multi-sensor IoT architecture	Improved monitoring accuracy	Limited scalability	Present project adds decision support
Xia et al., 2024	ML-based algal bloom prediction	Early bloom prediction	Needs large datasets	Present project uses threshold logic
Rahman et al., 2023	Low-cost ESP32 prototype	Affordable real-time monitoring	Small-scale only	Present project expands into full solution

3. ARCHITECTURE AND DESIGN OF THE PROPOSED FRAMEWORK

3.1 Design Overview

System design is a key phase of project development that defines the overall architecture, module interactions, and data flow within the proposed framework. It ensures effective integration of all components and provides a structured approach for achieving the project objectives. If the design is robust enough, the system will operate successfully, data will be processed efficiently, and the system will be reliable, and eventually maintainable.

In the IoT Based Non-Destructive Invasive Species Suppression System, it is intended to monitor an aquatic ecosystem in the form of a real environment in which automatic measures will be taken when any hint of imbalance is observed. The system uses a set of sensors, a microcontroller unit, logic for processing the sensors data and actuator modules to implement its core functionality.

The system design is based on a monitor – analyse – react concept. First , sensors monitor the habitat conditions. Then, the sensor's data is analyzed by ESP32 microcontroller, using a set of logic. Finally, once the system has detected imbalance in the measured parameters, preventive measures are taken to mitigate the invasion, while not affecting the native species.

3.2 Proposed Architecture

The system is designed to be modular, so that components can be added or removed as needed, without any major functionality issues. All the components have their unique role in the system, and they all work in synchronous manner to monitor the ecosystem.

The ESP32 microcontroller is the most important component. It is the part which bridges the sensor modules and the actuator modules. It receives signal from the sensors which includes a light sensor, a temperature and humidity sensor, gas sensor, and water level sensor. After the ESP32 receives these inputs, using them the ESP32 decides if the actuators need to act or not. The ESP32 also displays the data on the OLED screen, along with sending Whatsapp alerts to users. If the system detects a bad situation, it activates the actuators, which are pumps and fan to nullify the imbalance. Therefore, the IoT system helps to protect and maintain the ecosystem, and the combination of ESP32 and sensors is essential to enable the whole system to function.

The architecture can be proposed as follows:

- Sensor Module
- Signal Conditioning
- ESP32 Processing
- Monitoring Module
- Decision Support
- Actuator Module

All components are linked, so that the data can be passed from the time of collection to the time of taking action without any discrepancies. As an example, when the water becomes turbid, the ESP32 microcontroller evaluates the readings and concludes that there is a high risk thus it sends control signals to the relay module to turn on the pump to draw water from the source. Because of such modularity, the system is flexible and can be scaled up by adding additional functional modules.

As shown in the architectural flow of the system (Figure 3.1) the framework begins with the Sensor Module, where the DHT11, LDR, turbidity, MQ gas, and float sensors collect real-time environmental data. The sensor outputs undergo Signal Conditioning before being processed by the ESP32, which performs data normalization, risk calculation, and algae bloom risk classification. The processed information is displayed through the Monitoring Module using the OLED display and Serial Monitor. The Decision Support Module evaluates the risk level and generates alerts or control actions. Based on these decisions, the Actuator Module activates the pump, fan, turbulence pump, and buzzer to maintain favorable aquatic conditions and support ecosystem protection.

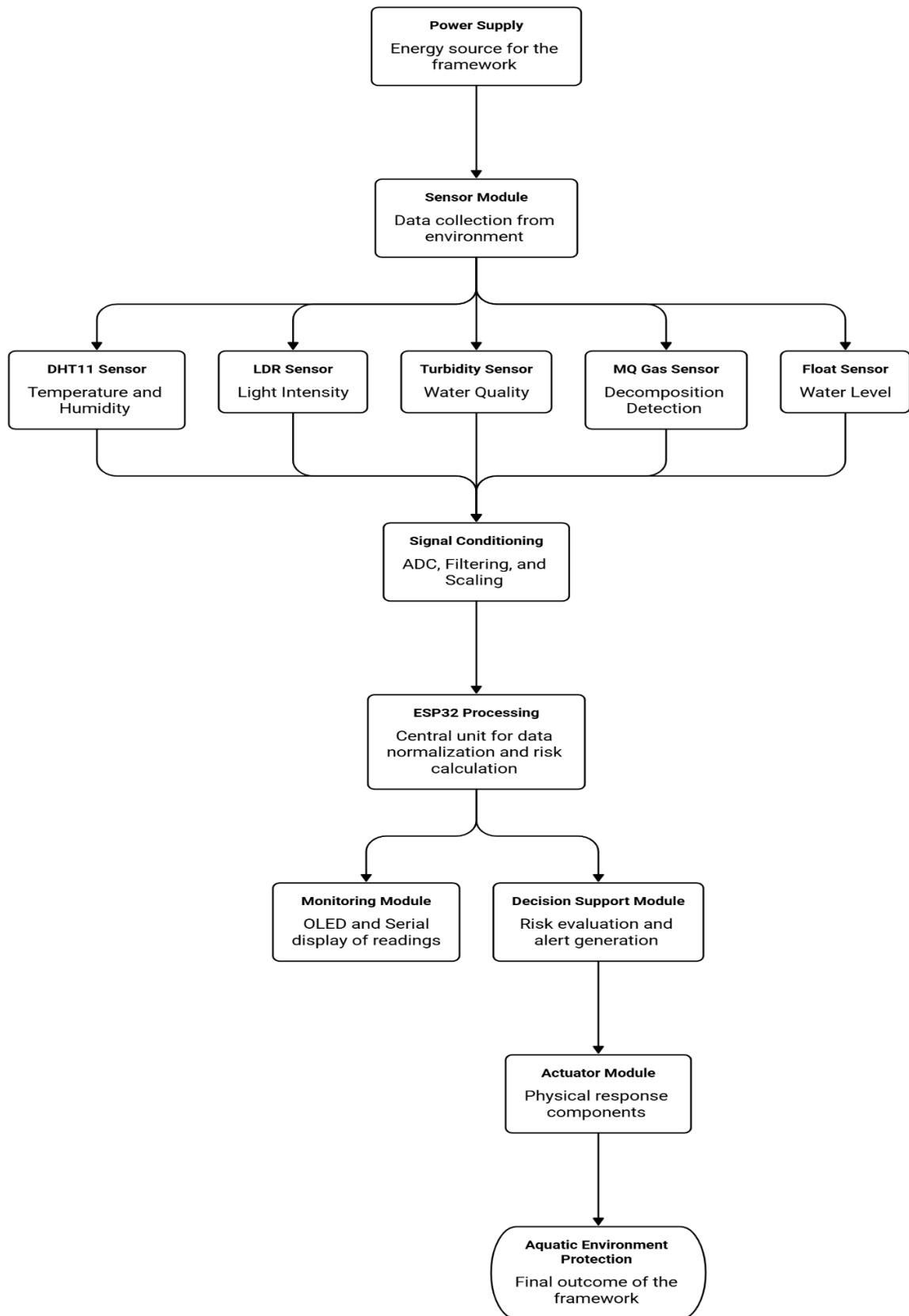


Figure 3.1: Architectural flow of the system

3.3 Functional Module Description

The system is split up into a number of modules to simplify construction and understanding, step by step.

1.Turbidity Sensor : The Turbidity sensor shown in Figure 3.3.1 is used to judge how clear the water is. The principle on which the sensor works is by measuring the suspended particles by light scattering, therefore more scattered light usually indicates higher turbidity. The sensor in this case does not enable a quantitative measurement, but rather assigns to the water one of the categories CLEAR, CLOUDY or HIGH turbidity, thus a quick judgement on the state of the water can be made. This is helpful to know the water quality and to detect conditions favorable for algal blooms, since higher turbidity is associated with higher organic content and more microbial activity. Turbidity levels change slowly thus continuous monitoring can be avoided to minimize power consumption.



Fig 3.3.1: Turbidity Sensor used for Water Quality Measurement

2.Light Sensor (LDR): LDR sensor shown in the figure 3.3.2 measures how much light actually lands on the water surface. Applications include things like plants covering the water surface, measurement of sunlight penetration, and identification of shading caused by invasive species. The light sensor is important, because algae rely on photosynthesis to grow. So when the sunlight exposure becomes strong, the algae growth speeds up. The turbidity sensors operate in relatively tough conditions in the water, the same cannot be said for temperature sensors. They tend to last longer, and their lifespan is between 2 to 5 years.



Fig 3.3.2: LDR sensor for Light Intensity Detection

3. Temperature and Humidity Sensor: Figure 3.3.3 shows the temperature sensor which checks the nearby ambient temperature around water body. Applications can be checking if temperatures are within the safe range , maintaining healthy habitat conditions. Algae tends to thrive when surroundings are warm. When the surrounding environment temperature goes up, the metabolic rate of algae also increases , so they reproduce faster and spread quickly. Humidity can also be used to make similar conclusions about the algal growth. Unlike turbidity sensors, temperature sensors are less subjected to challenging conditions in water environments, hence their longevity. Temperature sensors have a lifespan of between 2 to 5 years. Regarding coverage, temperature doesn't differ drastically within very small distances in open environments. Because of this, one sensor can cover an area of 1 to 3 meters radius, depending on where it is placed.



Fig 3.3.3: Temperature and Humidity Sensor for Environmental Monitoring

4. Gas Sensor (MQ-135): As shown in the Figure 3.3.4 The MQ-135 is a gas sensor designed to detect air quality around the ecosystem, and it can also sense other gases. Applications can be odor -based detection of organic matter, monitoring variations in the surrounding air. When algae bloom too much, they eventually die and start decomposing. Various gaseous substances such as carbon dioxide and ammonia are emitted. The MQ-135 sensor helps in detecting the effects of the algal bloom, especially when oxygen is depleted and the gases become prominent.



Fig 3.3.4: Gas Sensor for Air Quality Detection

It detects the quality of air in the area right above the water surface within a radius of

1 to 5 meters. This sensor uses a chemical sensing element that slowly decreases its performance, its expected lifespan is generally around 2 to 3 years.

5.Float Sensor: From the Figure 3.3.5 A float sensor basically measures the level of water that is present.

In practice it is used for preventing pumps from running dry, also to catch a shortage or even an overflow situation, and to keep overall safety inside the system. When the water level gets low, the concentration of nutrients within the water can go up. That ends up making a more favorable environment for algae growth. Also low water often also means there is less movement and more stagnation, so algae blooms get a better chance to happen. This type of sensor works at one fixed spot and checks whether the water level has reached a certain height. Since it is a simple mechanical device, it is durable and can last roughly 3 to 5 years, depending on how the water conditions are.



Fig 3.3.5: Float Sensor for Water Level Detection

6.OLED Display: Figure 3.3.6 shows OLED Display, where the system uses an SSD1306 OLED Display to show real time system information in a fairly simple, and clear format. It communicates through the I2C protocol so you only need two pins which makes it easy to interface with the ESP32. Rather than displaying unprocessed data, the system provides interpreted data such as light intensity, water

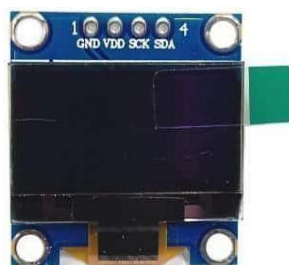


Fig 3.3.6: OLED display for displaying essential information

transparency, and the rate of decomposition of the material as well as the risk of algae growth. The system also displays the overall risk level. Additionally, the OLED displays the current state of actuators (fan/pump operation, etc.) and suggestions on how to decrease the risk levels. In case of low water volume, it changes to a special refill status screen. With such a design, the device can be installed in isolated systems due to its energy efficiency and clarity.

7.DC Fan: Figure 3.3.7 illustrates A DC fan, it is basically an electrical device that turns electrical energy into mechanical energy so it can push air around. source of power because of the nature of the electric current that runs to the motor. The electric current goes to the motor, where it rotates the blades. As the blades rotate, the movement forces air to circulate, hence circulation of air around the environment. For conventional systems, DC fans are employed to cool air. Algae develop faster when the conditions are warm. So when the temperature near the water body goes up, it forms a more suitable environment, for algae to multiply at a faster rate. When our system notices that the temperature has increased beyond the threshold, it turns the fan on automatically through the relay module that is connected to the ESP32. That helps lower the temperature a little and it also stops heat from building up.



Fig 3.3.7: DC Fan Used for Temperature Regulation

8.Water pump: Figure 3.3.8 shows a DC water pump which moves water from one place to another using electrical energy. It works by turning electrical energy into mechanical energy, in which a impeller or motor pushes water through pipes or a few outlet points. In the system the water pump matters a lot for reducing stagnation and keeping oxygen levels steadier, both of which are related to algae bloom formation. Algae often grows faster in still or stagnant water , because there is less disturbance. When the water is not moving, nutrients build up in the same location, sunlight

exposure stays almost the same, and oxygen levels can fall . The DC water pump helps by creating turbulence. It supports even distribution of nutrients , so algae does not gather in one spot. Also it enhances oxygen mixing in the water, which is important for a healthier aquatic environment

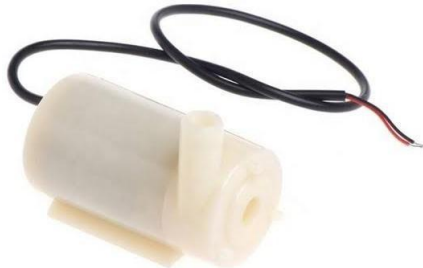


Fig 3.3.8: 5V Water Pump for Water Circulation and Pumping

9.ESP32 Microcontroller: As shown in the Figure 3.3.9, The ESP32 microcontroller works like the central processing unit of the whole setup, acting as the brain, that controls the operations . It gathers data from different sensors , then processes the data and send control signals to the relay to operate the actuators.ESP32 has advanced capabilities and is cost-effective, with high processing speed. It includes built in Wi-Fi and Bluetooth connections. It also has 32 general purpose input/output pins, which makes it easy to hold the sensors, actuators. Inside the system, the sensor information is sent over to the ESP32 first, for analysis and processing. Then, depending on the programmed rules, the ESP32 sends control signals to required actuator through the relay module , which helps the setup run more efficiently and in a more intelligent way. So overall it becomes a



Fig 3.3.9: ESP32 Development Module Used for Monitoring

dependable and adaptable option for embedded system building.

10.Decision Support Module: This module is made up of pre programmed logic that compares live sensor values against certain predetermined safe thresholds for risk classification, like Low Risk/Normal environmental conditions , Medium Risk/ Slight imbalance, requires monitoring and High Risk/ Strong probability for invasive dominance or poor water quality.

3.4 Software Requirements and Development Environment

The software tools used in this project are the Arduino IDE, which is used for writing, compiling, and uploading the program to the ESP32 microcontroller. The code is implemented using the Embedded C/C++ programming language. The Arduino IDE also provides a Serial Monitor for real-time observation and debugging of sensor data. Various software libraries such as WiFi, HTTPClient, DHT, Wire, Adafruit GFX, and Adafruit SSD1306 are utilized to support wireless communication, sensor interfacing, and OLED display functionality. Additionally, the software enables real-time data processing, risk assessment, actuator control, and WhatsApp-based alert notifications through the ESP32 platform.

3.5 Design Advantages

The system's design helps in achieving key goals such as low-cost implementation and real-time monitoring. The system is both efficient and portable and makes use of low-cost and beginner-friendly hardware, with also being an eco-friendly control approach. It is also scalable for larger systems and helps in protecting the native species from invasives by reducing manual effort. While designing the system, safety considerations include proper insulation of wires and the use of a low-voltage power supply that is safe. The electronics should be placed in waterproof conditions and prevention of pump dry run is done by using a float sensor.

The current design can be upgraded with a mobile application dashboard. It can also include cloud data storage, a solar power supply for this system, and a camera module that would help detect species. Along with that, an AI prediction model can also be added.

Summary of System Design:

The proposed design integrates multiple environmental sensors, an ESP32 microcontroller, monitoring interfaces, and automated actuator modules into a unified IoT framework. Sensor data is continuously collected, processed, and analyzed to assess algae bloom risk in real time. Based on the evaluated conditions, the framework provides alerts and activates appropriate control measures such as pumps, fans, and buzzers. The architecture offers a simple, cost-effective, and efficient approach for early detection of ecological imbalance and proactive aquatic environment protection. Table 3.1: Hardware Components and Their Functional Roles

S.NO	EQUIPMENT NAME	QUANTITY	PURPOSE
1.	Turbidity Sensor	1	Determines how cloudy or impure the water is.
2.	Light Sensor (LDR)	1	Measures how much light falls on the surface of water.
3.	Gas Sensor (MQ-135)	1	Detects air quality around the ecosystem
4.	Float Sensor	1	Measures the level of water present.
5.	Temperature Sensor	1	Measures the water temperature.
6.	ESP32 Microcontroller	1	Main controller of the system
7.	Relay Module	1	Activates the actuators.
8.	Water Pump	2	Supply of water
9.	Fan	1	For aeration.
10	OLED	1	Display the readings.
11.	7.4 Volts Battery	1	For power supply of ESP32
12.	9 Volts Battery	3	For power supply of each actuator.
13.	Buzzer	1	To act as an alarm.

Table 3.1 presents the hardware components utilized in the proposed framework along with their respective quantities and functions. The sensing unit comprises a turbidity sensor,

LDR sensor, MQ-135 gas sensor, float sensor, and temperature sensor for continuous monitoring of environmental conditions. An ESP32 microcontroller serves as the central processing unit, while a relay module interfaces the controller with the actuators. The control unit includes two water pumps, a fan, and a buzzer for automated intervention and alert generation. An OLED display provides real-time visualization of sensor readings and system status. Power is supplied through a 7.4 V battery for the ESP32 and separate 9 V batteries for the actuators, ensuring reliable operation of the entire framework.

4. DEVELOPMENT AND IMPLEMENTATION

4.1 Technologies Used

The proposed IoT-based aquatic monitoring framework integrates hardware, software, and communication technologies to enable real-time, non-destructive monitoring and automated environmental regulation. The framework employs low-cost sensing devices, wireless communication, and embedded processing to continuously assess aquatic conditions and support early intervention.

4.1.1 Hardware Technologies

ESP32 Microcontroller

- Serves as the central processing and control unit of the framework.
- Features a dual-core architecture with integrated Wi-Fi and Bluetooth connectivity.
- Responsible for sensor data acquisition, risk assessment, actuator control, and alert generation.

Sensors

- **MQ-135 Gas Sensor:** Monitors air quality and detects gases associated with decomposition and ecological imbalance.
- **LDR Sensor:** Measures ambient light intensity affecting algae growth.
- **Turbidity Sensor:** Evaluates water clarity and suspended particle concentration.
- **DHT11 Sensor:** Measures temperature and humidity for environmental monitoring.
- **Float Sensor:** Detects water level variations and supports automatic refill control.

Actuators

- **Water Pumps:** Provide water circulation and water-level maintenance.
- **DC Fan:** Supports aeration and temperature regulation.
- **Buzzer:** Generates audible alerts during critical conditions.

4.2.2 Software Technologies

Arduino IDE

- Used for Embedded C/C++ program development, compilation, debugging, and deployment to the ESP32.
- Provides the Serial Monitor for real-time observation of sensor readings and system status.

Software Libraries

- **WiFi Library:** Enables wireless connectivity.
- **HTTPClient Library:** Facilitates API-based communication.
- **DHT Library:** Supports temperature and humidity sensing.
- **Wire Library:** Enables I2C communication.
- **Adafruit GFX & SSD1306 Libraries:** Support OLED display operation.

WhatsApp API Integration

- Utilizes the UltraMsg API for real-time notification delivery.
- Sends alerts related to high-risk conditions, excessive sunlight exposure, and environmental recovery status

4.2 Hardware–Software Integration

System integration is achieved by incorporating the sensing, processing, communication, and control modules in a single integrated IoT platform for monitoring aquatic ecosystems status. The advantage of such integration is that it establishes a connectivity between all the hardware components and software components to ensure the transparency of the environment monitoring and the automatic reaction.

The processor and transmitter for the entire system is the ESP32 microcontroller. The microcontroller is utilized to connect to all of the sensors and actuators and is also responsible for running the logic (processing data and controlling the actuators based on the output of the sensors). It must be able to continuously gather environmental data, process it through the risk assessment calculation and trigger the appropriate response mechanisms if abnormal conditions are observed.

The sensors embedded into the sensing part of the system are capable of measuring several environmental parameters that affect the state of aquatic health and algal growth. The temperature and humidity are measured through DHT11 sensor as the microbial and algal activity can accelerate with the rise of temperature. LDR sensor that is built-in is used for measuring the intensity of sunlight as the increased exposure to sunlight facilitates the growth of photosynthesis and might result in a problem of algal bloom. The turbidity sensor is used for observing the water clarity as the increase of turbidity

can be caused by various suspended particles, microbial growth or a bloom inside the body of water. All of these imbalances affect the native species adversely.

Apart from this, the MQ-135 gas sensor is used to follow gaseous actions around the water surface area. The sensor is not measuring the concentration of an existing gas, rather it is used to monitor the degradation of organic matter or environmental changes. A float sensor is incorporated to check water level to maintain water conditions stable. When the water level is lower than the setpoint, water pump starts automatically and pumps the water back to the set point. The actuators include a refill pump, turbulence pump, cooling fan and buzzer.

The components are controlled via a relay module, which electrically isolates the low power ESP32 signals from the external power supply to the actuator. The actuators are powered by separate sources of battery power, so that the power required to drive a higher current device does not load the controller circuit. Local monitoring is achieved with an OLED display. Rather than continuous raw numbers being displayed, the display is updated to a more user friendly and less ambiguous format: LOW, MODERATE, HIGH, CLEAR, CLOUDY. This will greatly improve readability and usability of the endpoint for non technical users. The ESP32 has built in WiFi functionality and it is deployed to realize wireless communication. Under certain environmental conditions the controller will temporarily connect to the internet and send the user WhatsApp alerts. Warnings are sent for high bloom potential, high sun exposure conditions, and for recovery of normal conditions. During normal operation power usage is minimized by keeping WiFi off and only turning it on when alerts need to be sent. A strategy of 'grouped sensing' is used to reduce power consumption.

Parameters that change slowly are measured at longer intervals, such as turbidity and gas parameters. Parameters of importance such as water level are measured continuously. Such a method limits the number of sensor readings made and thus keeps the processors load down. The proposed system provides a compact, intelligent aquatic ecosystem monitoring system by incorporating environmental sensing, automatic control, wireless communication, and visualization at the local level, which can be used

for the early detection of algal blooms and management of water conditions.

4.3 Operational Workflow

The system begins to function instantly after plugging into power for the ESP32 and all the associated modules. With all of the controllers, sensors, relay outputs, OLED and communication methods powered by the battery, the system enters into a continuous monitoring state. That is, the system never actually stops running after completing the boot-up process. Monitoring and controlling activities are performed on a continuous and repeated basis thereby allowing for real-time monitoring of the environment, automatic online actions, and early warning support for managing the aquatic ecosystem.

With respect to the water level, the float sensor will always monitor the water level and whenever it is low, the refill pump will automatically start when the relay module engages and simultaneously provide a buzzer auditory alert that the water is being refilled because it is low. The group sampling intervals of this system will help to conserve power, which is an important factor.

Other parameters, specifically light intensity, will be checked at intervals using LDR sensor. The LDR readings provide insight as to how much light the surface of the water is receiving and how much sunlight is on average at that water's surface. The MQ-135 gas sensor monitors for gases that may be associated with the decomposition process around the aquatic area, and an increase in these readings would indicate that there is biological decomposition taking place in this aquatic area, as well as an increase in general strain on this aquatic ecosystem.

A turbidity sensor checks for signs of a decrease in water clarity within the aquatic area due to suspended solids, which may lead to accelerated algal growth in the aquatic area. Once all of the data collected from the sensors is received, the ESP32 will run the environmental risk model stored on it based on how each sensor collects data. Each of the parameters reported by the sensors will contribute to the calculation of the electrochemical risk score differently, based on their ecological relevance.

Once the sensor data is normalized and combined together, the probability of an algal

bloom or an ecosystem will be estimated. After the environmental risk score has been established, it will be captured and assigned to one of three categories: Low Risk, Medium Risk, and High Risk. When the environment is stable, no alerts for corrective action will be created or provided. However, when risk is present and increasing, then alerts will be generated via the OLED that are appropriate to the risk level and include guidance on appropriate corrective action.

When high-risk conditions are present, an automatic operation of the turbulence pump will commence to promote circulation of water in order to promote better oxygen mixing, and over time will eliminate stagnant water in the aquatic environment. In addition, when the temperature is above normal, the cooling fan will automatically turn on. Finally, the OLED will continuously update the user on all parameter readings related to their environment, including the status of all actuators and what action the user should take.

Instead of just displaying the raw parameter on the screen, the OLED will provide a summarized interpretation of the data (i.e., water quality, activity level, and light intensity) that is easier to read and provide the user with a quick understanding of the health of the ecosystem. The remote monitoring system sends critical alerts through WhatsApp only when they are significant so as to avoid excessive notifications. Monitoring and controlling activities are performed on a continuous and repeated basis thereby allowing for real-time monitoring of the environment, automatic online actions, and early warning support for managing the aquatic ecosystem.

4.4 Risk Assessment and Decision Logic

The proposed framework employs a risk assessment mechanism to evaluate the possibility of algae bloom formation using real-time environmental parameters. Sensor readings obtained from the light, temperature, turbidity, gas, and humidity sensors are processed by the ESP32 microcontroller. Since these parameters have different units and measurement ranges, they are first normalized and then combined using a weighted risk calculation model.

4.4.1 Data Normalization

To ensure uniform contribution from all environmental parameters, the sensor readings are normalized to a range between 0 and 1.

The normalized parameters are represented as:

- **L** = Normalized Light Intensity
- **T** = Normalized Temperature
- **Tu** = Normalized Turbidity
- **G** = Normalized Gas Level
- **H** = Normalized Humidity

Higher normalized values indicate environmental conditions that are more favorable for algae bloom development.

4.4.2 Weighted Risk Calculation

The overall risk score (**R**) is calculated using a weighted combination of the normalized parameters:

$$R = (0.25 \times L) + (0.20 \times T) + (0.30 \times Tu) + (0.15 \times G) + (0.10 \times H)$$

where:

Parameter Description		Weight
L	Light Intensity	0.25
T	Temperature	0.20
Tu	Turbidity	0.30
G	Gas Level	0.15
H	Humidity	0.10

Among all parameters, turbidity is assigned the highest weight (0.30) due to its strong association with water quality degradation and algae bloom conditions. Light intensity and temperature are also given significant importance because they directly influence photosynthetic activity and algae growth. Gas concentration and humidity provide supporting environmental information and therefore contribute smaller weights.

4.4.3 Risk Classification

Based on the calculated risk score, the environmental condition is classified into three categories:

Risk Score (R) Risk Level

$R < 0.30$ Low Risk

$0.30 \leq R < 0.60$ Medium Risk

$R \geq 0.60$ High Risk

- **Low Risk:** Indicates normal environmental conditions with minimal probability of algae bloom formation.
- **Medium Risk:** Indicates moderate environmental imbalance and the need for continuous monitoring.
- **High Risk:** Indicates a strong likelihood of algae bloom occurrence, triggering automated control actions and alert notifications.

The classified risk level is displayed on the OLED display and used by the control logic to activate the appropriate actuators and notification mechanisms.

5. RESULTS AND DISCUSSIONS

5.1 Results of Each Module

Different environmental conditions were applied to the developed IoT-based aquatic ecosystem monitoring system to evaluate the functionality and efficiency of the individual monitoring components. The testing process mainly focused on verifying sensor response, actuator performance, display functionality, wireless communication, and overall system reliability. Each module was tested individually, followed by an analysis of the complete integrated system operation.

Temperature, Humidity and Light Intensity Monitoring Module

The DHT11 and LDR sensors were used to continuously monitor temperature, humidity, and light intensity around the aquatic ecosystem. An artificial heat source was applied to increase the temperature beyond a predefined threshold, which triggered the ESP32 to activate a relay-controlled cooling fan automatically. The LDR sensor measured sunlight intensity under different lighting conditions. When excessive light was detected, the OLED display indicated a HIGH environmental condition and a WhatsApp alert was sent recommending the use of shade nets. Humidity data was also considered for environmental risk assessment. The system operated reliably throughout testing, providing stable monitoring, accurate sensor readings, automatic fan control, and timely alert generation for improved aquatic ecosystem management.

Water Turbidity and Decomposition Monitoring Module

The turbidity sensor and MQ-135 gas sensor were used to monitor water quality and decomposition activity within the aquatic ecosystem. The turbidity sensor was tested using clear water and water containing suspended particles, successfully classifying the conditions as CLEAR, CLOUDY, and HIGH turbidity. Higher turbidity levels significantly influenced environmental risk assessment by indicating the presence of suspended particles and possible biological activity. The MQ-135 sensor was used to detect gaseous activity near the water surface by exposing it to smoke and organic odor sources. Increased gas readings indicated higher decomposition activity and potential ecological imbalance. Experimental results showed that both sensors operated reliably under varying conditions, providing stable measurements and valuable inputs for environmental risk assessment and

algae bloom monitoring.

Water Level Monitoring, Refill and OLED Display Module

The water level monitoring system utilized a float sensor and an automatic refill mechanism to maintain the required water level within the aquatic ecosystem. During testing, different water levels were simulated to evaluate system performance. When the water level dropped below the preset threshold, the ESP32 activated a relay-controlled pump to refill the water automatically. Simultaneously, a buzzer alert and OLED warning message were triggered to notify the user. Once the water level returned to normal, the pump was automatically switched off. The OLED display provided real-time visualization of ecosystem conditions, including temperature, humidity, light intensity, water condition, decomposition activity, risk level, and actuator status. Experimental results showed reliable water level monitoring, successful automatic refill operation, accurate warning notifications, and clear display of environmental information, making the module effective for local monitoring and ecosystem management.

WhatsApp Alert and Automated Actuator Control Module

The WhatsApp alert module was developed to provide remote notifications during critical environmental conditions. Whenever the system detected a high algae bloom risk, excessive sunlight, or other abnormal conditions, the ESP32 enabled Wi-Fi connectivity and transmitted WhatsApp alerts through an integrated API. Recovery notifications were also sent when environmental conditions returned to normal, while alert flag logic prevented repeated messages for the same event. In addition, the automated actuator control module performed corrective actions based on sensor readings. The cooling fan was activated during high-temperature conditions, the turbulence pump operated under high-risk environmental states to improve water circulation and oxygen mixing, and the refill pump was triggered when low water levels were detected. Experimental results confirmed successful WhatsApp message delivery, stable wireless communication, reliable relay-based actuator operation, and effective automated environmental response, supporting efficient aquatic ecosystem management.

Overall System Performance Analysis

Compounded by the testing and evaluation carried out on each individual module, the sensing, environmental risk analysis, visualization, wireless communication and

automatic control modules of the proposed IoT-based aquatic ecosystem monitoring system were successfully realized, and the integrated system has achieved: stable monitoring and analysis function, reliable operation, sensor reliability, actuator control, and remote monitoring support, so the proposed complete aquatic ecosystem monitoring system was successfully implemented.

5.2 Objective-wise Results and Analysis

1.To continuously monitor key environmental parameters including turbidity, temperature, humidity, light intensity, gas concentration, and water level using integrated sensor modules.

The developed system successfully integrated multiple sensors for continuous monitoring of environmental conditions within the aquatic ecosystem. The DHT11 sensor accurately measured temperature and humidity, while the LDR sensor monitored sunlight intensity. The turbidity sensor effectively classified water conditions as clear, cloudy, or highly turbid. The MQ-135 sensor detected gaseous activity associated with decomposition processes, and the float sensor continuously monitored water levels. Experimental testing demonstrated stable sensor performance and reliable data acquisition under different environmental conditions, providing comprehensive real-time ecosystem monitoring.



Figure 5.2.1 - OLED Screen Displaying Environmental Sensor Data

2.To implement automated environmental control using relay-driven actuators such as water pumps, turbulence pumps, cooling fans, and buzzer alerts based on detected risk conditions.

The automated control module responded effectively to abnormal environmental conditions. When temperature exceeded predefined thresholds, the cooling fan was activated automatically. High-risk environmental conditions triggered the turbulence pump to improve water circulation and oxygen mixing, while low water levels activated the refill pump and generated buzzer alerts. Relay-controlled actuators operated reliably throughout testing and responded according to the programmed

environmental control logic. The results confirmed successful automated intervention and reduced dependence on manual ecosystem management.

3. To develop a real-time risk assessment model that processes sensor data and classifies ecosystem conditions into LOW, MEDIUM, and HIGH risk levels.

A weighted environmental risk assessment model was implemented to evaluate ecosystem conditions based on multiple environmental parameters. Sensor data from temperature, humidity, turbidity, light intensity, and decomposition activity were analyzed collectively to calculate an overall risk score. The system successfully categorized environmental conditions into LOW, MEDIUM, and HIGH risk levels. Experimental results showed that the multi-parameter approach improved assessment reliability compared to single-parameter monitoring systems and provided effective early detection of conditions favorable for algal bloom development.



Figure 5.2.2 - OLED Screen Displaying Environmental Sensor Data

4. To provide real-time user notifications through a WhatsApp alert system for critical environmental conditions and early intervention.

The wireless communication module successfully delivered real-time WhatsApp notifications whenever critical environmental conditions were detected. Alerts were generated for high bloom risk, excessive sunlight exposure, and restoration of safe environmental conditions. The alert flag mechanism prevented redundant notifications during prolonged abnormal states. Testing confirmed stable Wi-Fi communication and successful message transmission, enabling remote monitoring and timely intervention without requiring continuous physical supervision.

5.To develop a sustainable, non-chemical, and user-friendly solution for aquatic ecosystem protection through continuous monitoring, automated control, and early warning mechanisms.

The proposed system successfully demonstrated a sustainable and environmentally friendly approach to aquatic ecosystem management. By combining continuous sensor monitoring, automated actuator control, OLED-based visualization, and remote WhatsApp alerts, the system minimized the need for chemical treatments and manual interventions. The integrated platform operated reliably under various environmental conditions while maintaining low power consumption and cost effectiveness. The overall results indicate that the system can support early algal bloom prevention, improve water quality management, and provide a practical solution for protecting aquatic ecosystems.



Figure 5.2.3: Working Prototype of IoT-Based Invasive Species Control System



Figure 5.3 - Experimental Setup for Monitoring Aquatic Ecosystems

In figure 5.3, we see the experimental setup used to monitor an aquatic ecosystem environment by observing different environmental parameters in real-time. This setup contains both aquatic plants and equipment for observing ecosystem condition and changes in algae levels.



Figure 5.4 - OLED Screen Displaying Environmental Sensor Data

In figure 5.4, we see a real-time display of data collected by various environment sensors that are connected to the monitoring system on an OLED screen. The data that is displayed for analysis and monitoring purposes includes temperature, light levels, turbidity, gases, and amount of water.



Figure 5.5: Working Prototype of IoT-Based Invasive Species Control System

Figure 5.5 shows a working prototype of an IoT based Invasive Species Control System integrated with sensors and control modules, demonstrating its use in Monitoring environmental conditions and performing automating of Functions to help support aquatic ecosystem stability.

6. CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The proposed IoT-based aquatic ecosystem monitoring system successfully demonstrated the effectiveness of using modern IoT technologies for sustainable freshwater ecosystem management. By integrating multiple sensors for monitoring temperature, humidity, turbidity, light intensity, gas concentration, and water level, the system provided continuous real-time environmental monitoring and accurate ecosystem assessment. The implementation of a risk assessment model enabled early detection of unfavorable environmental conditions that could lead to algal blooms and water quality degradation. Compared to conventional monitoring approaches, the developed system offered faster response, improved monitoring efficiency, and reliable environmental condition evaluation.

In addition, the system incorporated automated actuator control and WhatsApp-based alert mechanisms to support proactive ecosystem management. Automated devices such as cooling fans, turbulence pumps, and refill pumps responded effectively to abnormal environmental conditions, reducing the need for constant human intervention. The use of non-chemical control methods promoted environmentally sustainable ecosystem protection while supporting responsible water resource management. Furthermore, real-time notifications enhanced user awareness and enabled timely preventive actions. Overall, the proposed system provides a low-cost, reliable, and scalable solution for aquatic ecosystem monitoring, contributing to freshwater conservation and supporting Sustainable Development Goal 6 (Clean Water and Sanitation).

6.2 Future Enhancements

Future enhancements of the proposed system can include the integration of Artificial Intelligence (AI) and Machine Learning (ML) techniques for predictive algal bloom detection. By utilizing historical environmental sensor data, AI models can analyze patterns and forecast algal bloom events before they occur, enabling proactive ecosystem management. In addition, the system can be expanded by incorporating advanced sensors

for measuring nitrate, phosphate, dissolved oxygen, pH, and heavy metal concentrations. This enhancement would improve the detection of agricultural runoff, industrial pollutants, and chemical contamination, resulting in a more comprehensive assessment of water quality.

Another potential improvement is the integration of renewable energy sources, such as solar panels, to make the system energy-independent and suitable for deployment in remote aquatic environments. This would reduce dependence on external power supplies and enhance long-term operational sustainability. Furthermore, a dedicated mobile application can be developed to provide real-time monitoring, alert notifications, historical data visualization, and remote system control. Such a platform would enable farmers, environmental agencies, and local authorities to conveniently monitor and manage ecosystem conditions through smartphones or tablets.

GLOSSARY

1. Internet of Things (IoT)

A system of devices that sense and transmit information with each other without the need for human intervention.

2. ESP32 Microcontroller

A low-cost, high-performance microcontroller with built-in Wi-Fi and Bluetooth used to process data and control devices in IoT applications.

3. Turbidity

A measure of water clarity that indicates the presence of suspended particles such as dirt, algae, or pollutants.

4. LDR (Light Dependent Resistor)

A sensor that measures light intensity based on resistance variation when exposed to light.

5. DHT Sensor

A temperature and humidity sensor used to monitor environmental conditions around the system.

6. MQ-135 Gas Sensor

A sensor used to detect harmful gases such as ammonia, carbon dioxide, and air pollutants.

7. Relay Module

An electronic switch that allows a microcontroller to control high-power devices like pumps or fans.

8. Signal Conditioning

The process of converting raw sensor signals into stable and usable data for processing.

9. Embedded System

A combination of hardware and software designed to perform a specific function within a larger system.

10. Real-Time Monitoring

Continuous observation and data collection of environmental conditions without delay.

11. Actuator

A device that performs an action based on signals from the controller, such as turning on a pump or fan.

12. Ecological Balance

A state of equilibrium in which there is no disturbance between living things and environmental factors.

APPENDIX

Appendix A: Hardware Components

- ESP32 Microcontroller
- Turbidity Sensor
- LDR Sensor
- Temperature&Humidity Sensor
- MQ-135 Gas Sensor
- Relay Module
- Connecting wires
- Breadboard

• Powe
r supply (5V
power bank)

Appendix B:

Software

Tools

- Arduino IDE

• Embedd
ed C / C++
programming

Appendix C:

Working Setup

The system is set up in a controlled environment with sensors around the water body. The ESP32 is responsible to collect data and show the data through the serial monitor.

Appendix D: Sample Output

- Low Risk: Normal turbidity and sufficient light penetration
- Medium Risk: Slight increase in turbidity and reduced light
- High Risk: High turbidity, poor light, and possible ecological imbalance

Appendix E:

Limitations of Prototype

- Small-scale model
- No cloud integration
- Limited dataset for prediction
- Manual setting of threshold values

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Important: Since the numbering changed, all in-text citations in Chapters 1 and 2 must also be updated to match this final numbering. This is now the correct strictly alphabetical reference order.